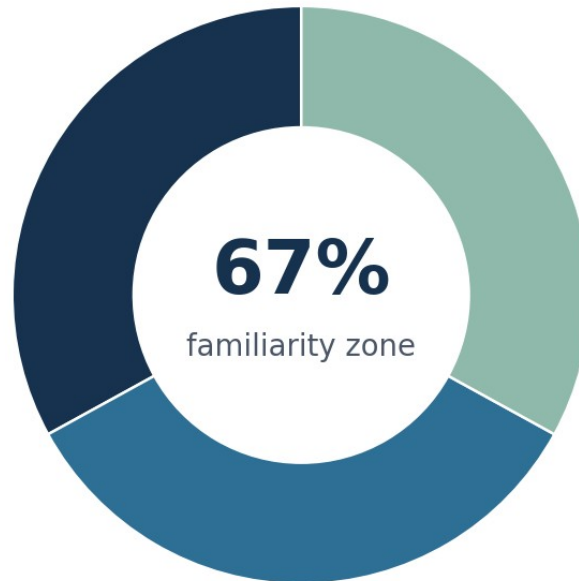


MATH 2123

COMPLETE BACKLOG PREPARATION GUIDE

Repeat-first strategy • Same-type mastery • Regular-question survival



Exact repeat - 33% Same type - 34% Regular / new - 33%

PERSONAL BACKLOG MODEL 33% exact repeat + 34% same type + 33% regular/new. This guide treats that split as your planning assumption, not as a statistically verified rate from the uploaded papers.

Made by 2223033

Built from the uploaded KUET Math 2123 term-question collection, topic-wise solved compilations, lecture notes and reference material.

How to Use This Guide

A backlog-safe system built around familiarity, method recognition and controlled risk.

What this guide is - and is not

This is a preparation guide, not a prediction guarantee. The uploaded source set is dominated by regular term-final papers and topic-wise compilations. No separate backlog-paper archive was supplied. Therefore, the historical question evidence comes from the regular Math 2123 bank, while the 33% / 34% / 33% split is your personal backlog assumption.

KEY CONSEQUENCE If your 33/34/33 model is approximately right, 67% of the paper lives in a familiarity zone: either an exact repeat or a recognizable template. The backlog strategy should therefore maximize recall + template recognition before spending large time on rare, novel variations.

The 4-layer preparation stack

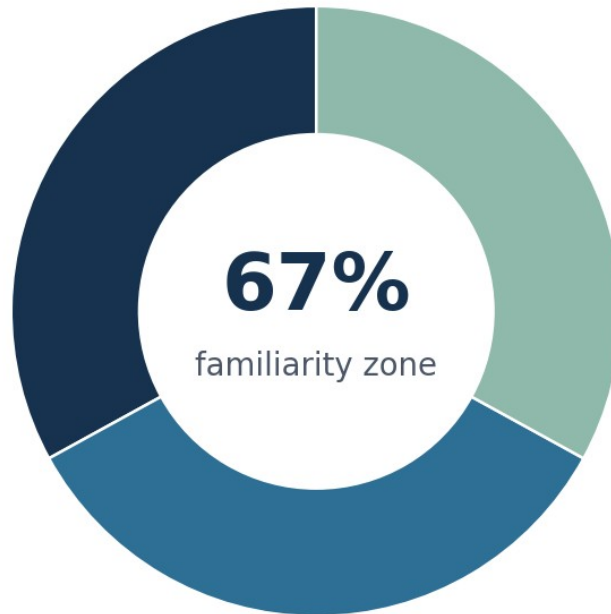
Layer	Goal	What you do	Target
1. Exact-repeat bank	Recall without thinking	Solve/rewrite repeated questions until the steps are automatic	100% reproducible
2. Same-type template bank	Adapt under changed numbers	Learn method triggers, setup and mutation patterns	Recognize in < 90 sec
3. Regular/new coverage	Avoid blank answers	Know formulas, definitions, theorem selection and first steps	Salvage + partial/full marks
4. Exam execution	Convert knowledge into marks	Choose 3 of 4 sets per section using confidence and time	No emotional question selection

One rule throughout the guide

3 + 1 RULE You must answer 3 sets from each section out of 4. Prepare 3 sets to mastery and the 4th as an insurance set. Never enter a backlog exam with exactly 3 prepared sets; a single unusual question can destroy the plan.

The 33 / 34 / 33 Model

Turn the personal assumption into a rational study-time plan.



Exact repeat - 33% Same type - 34% Regular / new - 33%

Probability is not the same as study-time allocation

Exact repeats are cheaper to prepare than same-type questions. Same-type questions require method understanding, so they deserve slightly more study time than their probability share. Regular/new questions receive enough coverage to prevent blank answers, but they should not consume the majority of backlog preparation time.

Preparation bucket	Paper assumption	Recommended study time	Why
Exact repeat	33%	35%	High return: memorized structure + known final route.
Same type	34%	40%	Highest adaptability demand; one template can cover many variants.
Regular / new	33%	25%	Cover fundamentals and salvage strategy rather than chasing every possible problem.

How to tag every question in your bank

Tag	Meaning	Study action
R1 - Repeat	Exact or near-exact historical repeat	Memorize clean solution path; timed

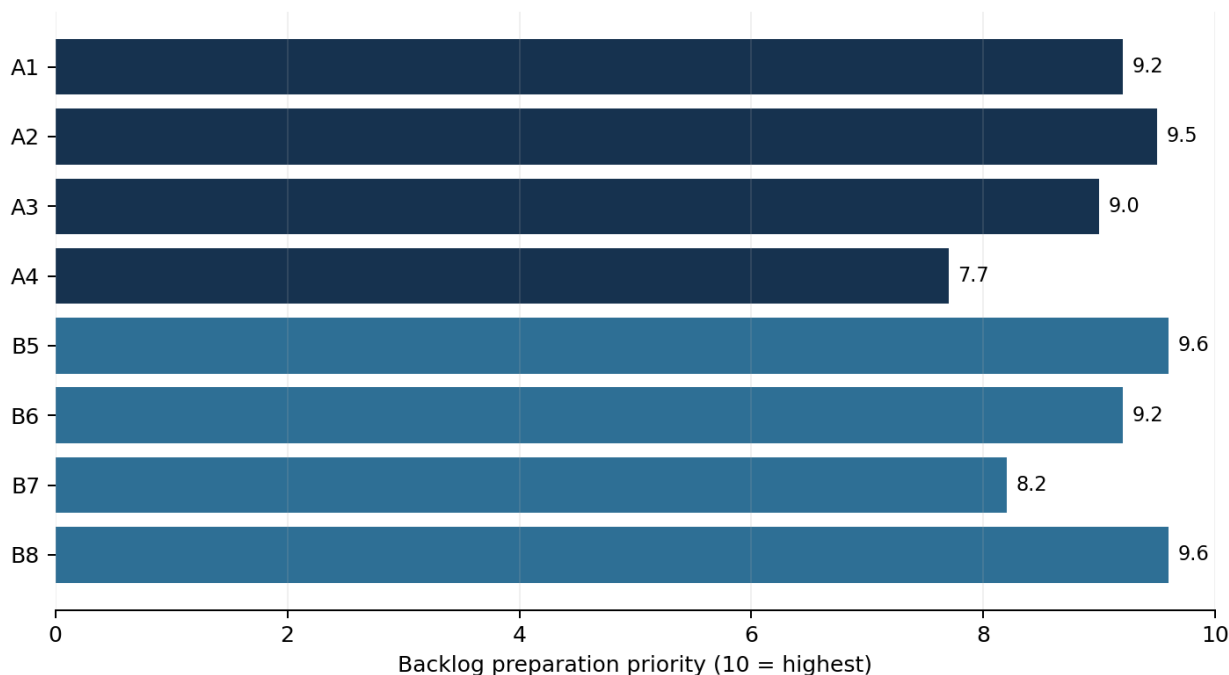
Tag	Meaning	Study action
		reproduction.
R2 - Template	Same mathematical structure, changed data/path/matrix/load	Write the trigger + 5-step method; drill variants.
R3 - Regular	No strong repetition signal	Know concept, formula, setup and first 40-60% of solution.

BACKLOG MINDSET A difficult new question is allowed to exist. Your job is not to make every question easy; your job is to make at least 3 sets in each section reliably answerable.

SET DISTRIBUTION

Eight Preparation Sets

These are preparation buckets; historical papers sometimes move topics between Section A and Section B.



Set	Preparation bucket	Repeat character	Priority	Minimum backlog outcome
A1	Matrix fundamentals, types, properties, short proofs	Theory repeats very strongly	Very High	Definitions + decomposition/properties secure
A2	Inverse, rank, consistency, dependence / independence	Same-type numerical templates	Very High	Can classify and solve systems confidently
A3	Echelon/canonical/normal form, eigenvalues, Cayley-Hamilton	Workflow repeats	Very High	One full matrix transformation + one eigen/C-H route
A4	Vector kinematics, mechanics, geometry	Method-based; moderate variability	High	Velocity/acceleration + work/geometry basics
B5	Direct Laplace, periodic, piecewise, unit step	Exact + template repeats	Very High	Periodic/unit-step/direct transform automatic
B6	Inverse Laplace, convolution, ODE/integral equation, beam	Template-heavy	Very High	Convolution + ODE + one beam family
B7	Gradient, directional derivative, div/curl, tangent/normal	Concept + formula template	High	Operator use + physical meaning + tangent plane
B8	Line, surface, volume integration; Green/Stokes/Gauss	Exact + same-type repeats	Very High	Correct theorem and limits, no setup errors

BEST GENERAL BACKLOG CONFIGURATION Section A: master A1 + A2 + A3, keep A4 as insurance. Section B: master B5 + B8, then choose B6 or B7 as your third core set based on personal strength; keep the other as insurance.

SECTION A

Preparation Map

Master 3 sets, prepare the 4th as an insurance route.

Section A selection logic

Profile	Core sets	Insurance	Reason
Matrix-strong	A1 + A2 + A3	A4	Highest repeat/template concentration.
Vector-comfortable	A1 + A2 + A4	A3	Use vector methods if eigen/normal-form work is slow.
Very short preparation	A1 + A2 + A3	A4 definitions only	Matrix questions are more procedural and easier to checklist.

Section A non-negotiables

- Never confuse symmetric ($A^T = A$), skew-symmetric ($A^T = -A$), orthogonal ($A^T A = I$), singular ($\det A = 0$), idempotent ($A^2 = A$), involutory ($A^2 = I$).
- For consistency questions, always compare $\text{rank}(A)$, $\text{rank}([A|B])$ and the number of unknowns; do not decide by intuition.
- For eigenvalues, write $\det(A - \lambda I) = 0$ clearly before solving. For eigenvectors, substitute each λ into $(A - \lambda I)X = 0$.
- For Cayley-Hamilton, first obtain the characteristic polynomial correctly; a sign error destroys the rest.
- For vector motion, differentiate the position vector componentwise; speed is $|v|$, not v .

SECTION A GOAL Have three complete set scripts that you can finish even under stress, plus one rescue set whose definitions/formulas can still earn partial marks.

Matrix Fundamentals & Repeat-Theory Bank

High-return definitions, properties and short proofs.

Why A1 is backlog gold

The theory compilation shows repeated matrix definitions across many years. Orthogonal matrix alone appears in multiple papers, while singular matrix, rank, symmetric/skew-symmetric and eigen concepts recur repeatedly. This is the cheapest source of reliable marks because the wording changes less than long numerical problems.

Priority	Question / concept	Evidence in uploaded bank	Backlog action
★★★★★	Orthogonal matrix + property $A^{-1} = A^T$	Appears repeatedly across 2016, 2018, 2019, 2022, 2023, 2024	Memorize definition, verification method and direction-cosine proof.
★★★★★	Symmetric + skew-symmetric decomposition	2017 & 2018 proof family	Know $A = \frac{1}{2}(A+A^T) + \frac{1}{2}(A-A^T)$, including uniqueness idea.
★★★★☆	Singular / non-singular matrix	Repeated definition family	Definition + determinant test + one 2×2 example.
★★★★☆	Rank of matrix	Repeated 2019, 2023, 2024 definition family	Definition + number of non-zero echelon rows.
★★★★☆	Eigenvalue / eigenvector definition	Repeated theory family	Definition + characteristic equation.
★★★☆☆	Diagonal, elementary, Hermitian, idempotent, triangular	Rotating definition bank	Prepare 1-line definition + example each.

A1 exact-answer template

1. Write the definition in mathematical form.
2. Give the minimum correct condition (e.g., $ATA = I$).
3. Add one clean example if asked.
4. For proof questions, use the definition immediately; avoid narrative filler.

A1 45-minute drill

Minutes	Task
0-10	Write 10 matrix definitions from memory.
10-20	Prove diagonal entries of a skew-symmetric matrix are zero; prove inverse of symmetric non-singular matrix is symmetric.
20-35	Do one symmetric + skew-symmetric decomposition numerical problem.
35-45	Verify one matrix is/is not orthogonal; check answer presentation.

PASS-SAVING DETAIL If a proof is forgotten, start from the defining equation. Example: skew-symmetric means $a_{ij} = -a_{ji}$; set $i = j$ to immediately get $a_{ii} = 0$.

Inverse, Rank, Consistency & Linear Dependence

The strongest same-type matrix workflow set.

Recognition triggers

Question wording	Immediate method
"Find inverse using elementary transformation"	Augment $[A \mid I]$ and row-reduce to $[I \mid A^{-1}]$. First check whether inversion is possible.
"Find rank"	Convert to echelon/normal form; count non-zero rows / pivots.
"For what value of k is the system consistent / unique / infinite / no solution?"	Row-reduce $[A B]$, then compare ranks and unknown count.
"Linearly dependent or independent?"	Use determinant (square set) or rank / non-trivial combination.
"Solve by matrix method"	Form $AX=B$, then use inverse/elimination as suitable.

The consistency decision table

Condition	Meaning	Conclusion
$\text{rank}(A) \neq \text{rank}([A B])$	Contradictory augmented row appears	No solution / inconsistent
$\text{rank}(A) = \text{rank}([A B]) = n$	Pivot for every unknown	Unique solution
$\text{rank}(A) = \text{rank}([A B]) < n$	Free variable(s)	Infinitely many solutions

Same-type mutation patterns

- Parameter k moves from a coefficient to the right-hand side, but the rank test is unchanged.
- A 3×3 inverse becomes a 4×4 inverse: the process is the same, only arithmetic grows.
- Linear-dependence questions may be written as vectors, rows of a matrix or homogeneous equations; convert them to one consistent framework.
- Network-flow style equations are still linear systems: define unknowns carefully before elimination.

COMMON BACKLOG ERROR Do not continue an inverse calculation after a pivot collapse proves the matrix is singular. State clearly that A^{-1} does not exist and justify it by $\det(A)=0$ or $\text{rank}(A)<n$.

Normal Form, Eigenvalues & Cayley-Hamilton

Procedural, high-value, and very trainable.

A3 core workflow 1 - Echelon → Canonical → Normal form

1. Use elementary row operations to create pivots and zeros below them.
2. Continue to reduced/canonical structure as required.
3. If column operations are allowed, reduce to the normal form $\text{diag}(I_r, 0)$.
4. Rank = r , the number of pivots / identity entries.
5. If asked for P and Q such that PAQ is normal form, track operations or construct the required elementary products carefully.

A3 core workflow 2 - Eigenvalue → Eigenvector

1. Form $\det(A - \lambda I) = 0$.
2. Solve the characteristic equation for λ .
3. For each eigenvalue, solve $(A - \lambda I)X = 0$.
4. Choose a convenient non-zero scalar multiple for the eigenvector.
5. Use trace/determinant checks when practical: sum of eigenvalues = $\text{trace}(A)$; product = $\det(A)$.

A3 core workflow 3 - Cayley-Hamilton

C-H TEMPLATE If $p(\lambda) = \lambda^n + c_1\lambda^{n-1} + \dots + c_n$ is the characteristic polynomial, then $p(A) = 0$. Use this identity to reduce high powers of A or express A^{-1} when A is non-singular.

Same-type mutation patterns

- Different numerical matrix; identical eigen workflow.
- “Largest eigenvalue” only changes which eigenvector you finally report.
- A power such as A^5 or A^{-1} appears; reduce it using the characteristic identity.
- Normal-form questions change matrix size but still depend on systematic elementary operations.

Vector Kinematics, Mechanics & Geometry

Your Section A insurance set - or a core set if vectors are stronger than matrices.

Fast formula bank

Topic	Formula / trigger	What examiners usually change
Motion	$v = dr/dt$, $a = dv/dt$, speed = $ v $	Position vector, evaluation time, direction.
Unit tangent	$T = v/ v $	Curve and point/time.
Work done	$W = \int_C F \cdot dr$	Path: direct, parametric, piecewise, curve relation.
Direction cosine / angle	$\cos\theta = (a \cdot b) / (a b)$	Vectors/surfaces.
Scalar triple product	$a \cdot (b \times c)$	Volume / coplanarity interpretation.
Vector triple product	$a \times (b \times c) = b(a \cdot c) - c(a \cdot b)$	Proof + identity manipulation.

A4 backlog method

1. Parameterize first; do not integrate in mixed variables.
2. Write dr explicitly before $F \cdot dr$.
3. For motion, evaluate derivatives before substituting the requested time unless simplification is obvious.
4. When asked for physical/geometrical meaning, give both the formula and one sentence of interpretation.

INSURANCE SET RULE If A4 is only your insurance set, prioritize motion + work done + scalar/vector triple-product theory. These are more reusable than rare geometric subcases.

SECTION B

Preparation Map

Two high-confidence anchors (B5, B8) + one chosen core (B6 or B7) + one insurance.

Choose your third core set

If you are better at...	Choose as core	Keep as insurance	Why
Algebra / transforms	B6	B7	Inverse Laplace, convolution and ODEs are procedural.
Vector calculus / geometry	B7	B6	Gradient, directional derivative and div/curl rely on compact formulas.
Unsure	B6 first, B7 second	Whichever is slower after 2 mocks	B6 also covers beam/ODE families, so it has broader payoff.

Section B anchors

B5 ANCHOR Periodic / piecewise / unit-step Laplace has strong repetition and very stable formulas.

B8 ANCHOR Line/surface/volume integration has repeated exact questions and repeated theorem-selection patterns. Setup accuracy is the main scoring skill.

Direct Laplace, Periodic, Piecewise & Unit Step

One of the highest-return backlog sets.

Exact-repeat signal

The uploaded unit-step compilation identifies the same periodic piecewise function as repeated in 2021, 2022 and 2024: $F(t)=3t$ for $0<t<2$, $F(t)=6$ for $2<t<4$, with period 4. This is a model example of why periodic-function technique belongs in the exact-repeat bank.

Question family	Trigger formula	Backlog status
Periodic function	$L\{f\} = [1/(1-e^{-sT})] \int_0^T e^{-st} f(t) dt$	R1/R2 - must master
Unit step	$u(t-a)=0$ before a , 1 after a	R1/R2 - must master
Second shifting	$L\{f(t-a)u(t-a)\}=e^{-as}F(s)$	R1/R2 - must master
Piecewise direct definition	Split $\int_0^\infty$ at break points	R2 - very common
Multiplication by t^n	$L\{t^n f(t)\}=(-1)^n d^n F/ds^n$	R2 - formula based
Derivative transform	$L\{f^{(n)}\}=s^n F - s^{n-1}f(0)-\dots$	R2/R3

Same-type mutation checklist

- Period 4 becomes 2π or another T .
- Breakpoints 2 and 4 become a and b .
- A delayed sine/cosine replaces a delayed polynomial.
- A graph is given instead of a formula; read intervals carefully.
- The question asks you to express the function in unit-step form before transforming it.

B5 SCORING RULE If the function is periodic, write the periodic-transform formula before doing any integration. If it is delayed, rewrite around $(t-a)$ before using the shift theorem.

Inverse Laplace, Convolution, ODEs & Beam Deflection

Template-heavy; preparation depth pays off.

Method-selection table

Expression / problem	Best first method
Rational $F(s)$, easy factorization	Partial fractions + standard inverse table
Product $F(s)G(s)$	Convolution theorem if partial fractions are awkward
Shifted denominator $(s-a)^2+b^2$	First shifting / complete the square
Initial-value ODE	Transform derivatives, substitute initial values, solve $Y(s)$, invert
Integral equation	Transform both sides; use $L\{\int_0^t f(u)du\} = F(s)/s$
Beam deflection	El $y^{(4)}=w(x)$; encode load with unit step, transform, apply boundary conditions

Beam family - same type, different support/load

Historical family	Support / load type	What to memorize
2017	Cantilever, discontinuous UDL	Clamped: $y=y'=0$; free: $y''=y'''=0$
2018	Clamped-clamped, discontinuous triangular load	Both ends: $y=y'=0$
2019	Clamped-simply supported, discontinuous UDL	Left $y=y'=0$; right $y=y''=0$
2022	Simply supported, continuous UDL	Both ends: $y=y''=0$

BEAM SURVIVAL FORMULA Clamped/embedded: $y=0, y'=0$. Simply supported/hinged: $y=0, y''=0$. Free end: $y''=0, y'''=0$. These boundary conditions are more important than memorizing one final deflection expression.

B6 minimum viable mastery

- 2 inverse-Laplace partial-fraction problems
- 2 convolution problems
- 2 initial-value ODEs
- 1 integral equation
- 1 discontinuous-load beam + 1 simply supported beam

Vector Differential Calculus

Compact formulas; strong rescue potential.

Core operator table

Operation	Definition	Question trigger
Gradient	$\nabla\phi = \phi_x \mathbf{i} + \phi_y \mathbf{j} + \phi_z \mathbf{k}$	Normal to level surface; maximum directional derivative
Directional derivative	$D_u\phi = \nabla\phi \cdot \mathbf{u}$	"In the direction of..."
Divergence	$\nabla \cdot \mathbf{F} = P_x + Q_y + R_z$	Solenoidal / source-sink
Curl	$\nabla \times \mathbf{F}$	Irrotational / rotation
Laplacian	$\nabla^2\phi = \phi_{xx} + \phi_{yy} + \phi_{zz}$	Harmonic potential / identity
Tangent plane	$\nabla F(\mathbf{P}) \cdot [\mathbf{x}, \mathbf{y}, \mathbf{z}] - \mathbf{P} = 0$	Surface $F(x,y,z)=0$ at a point

Physical meaning - easy theory marks

- Gradient points in the direction of greatest increase and is normal to the level surface.
- Divergence measures net outward flux density; zero divergence means solenoidal.
- Curl measures local rotation; zero curl means irrotational under suitable domain conditions.
- The maximum directional derivative is $|\nabla\phi|$ and occurs along $\nabla\phi$.

Same-type mutations

- Different scalar field and point; same gradient/directional-derivative workflow.
- Surface equation changes; tangent-plane formula stays identical.
- Parameters a, b, c are chosen to make a vector field solenoidal or irrotational; set coefficients of divergence/curl to zero.
- Potential-function problems ask you to integrate components consistently and reconcile constants.

B7 RESCUE VALUE Even when arithmetic becomes messy, correct ∇ , divergence/curl setup, unit direction vector and tangent-plane formula can preserve substantial partial credit.

Line, Surface & Volume Integration

A major repeat zone - but setup errors are fatal.

Exact / near-exact repeats documented in the uploaded integration compilation

Repeat family	Years	What repeats
Piecewise vs direct line integral	2016 & 2017	Same vector field/path family.
Work on curve $x^2=4y$, $3x^3=8z$	2020 & 2024	Same force-field/curve question family.
Cylinder surface flux $x^2+y^2=16$, first octant, $0 \leq z \leq 5$	2018 & 2023	Same surface-flux family.
Volume integral of $\text{div } \mathbf{F}$ on cube $1 \leq x, y, z \leq 3$	2022 & 2023	Same volume-divergence family.
Green theorem triangle/circle	Multiple years	Different P,Q and region; same conversion template.

Theorem selection in 15 seconds

You see...	Think...	Check
Closed planar line integral $\oint \mathbf{P}dx + \mathbf{Q}dy$	Green	Orientation positive/CCW; region simple?
$\oint \mathbf{C} \cdot d\mathbf{r}$ and $\mathbf{C}$ bounds a surface	Stokes	Curl easier than direct line integral?
Closed surface flux $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$	Gauss	Surface genuinely closed?
Open surface flux	Direct surface integral	Choose projection / parameterization.
Work along a specified open path	Direct line integral	Parameterize each segment/curve.

B8 error-control checklist

- Write the region limits before integrating.
- For cylindrical coordinates, include the Jacobian r in dV .
- For sphere/cylinder surface elements, do not use volume Jacobians by mistake.
- Check outward vs inward normal and path orientation.
- If a path is piecewise, split it explicitly and add segment results.

HIGH-YIELD PRACTICE Solve every repeated B8 family once slowly, then twice under time. After that, practice changed geometry rather than repeating identical arithmetic.

Exact / Near-Exact Repeat Priority

Ranked by repetition evidence and backlog usefulness.

Rank	Question family	Repeat evidence	Set	Required mastery
1	Orthogonal matrix definition / verification / proof family	Multiple papers (2016, 2018, 2019, 2022, 2023, 2024)	A1	Word-perfect definition + verification + direction-cosine proof
2	Periodic piecewise Laplace: $3t/6$, period 4	2021, 2022, 2024	B5	Full transform from formula without notes
3	Rank definition / rank method	2019, 2023, 2024 theory family	A1/A2	Definition + echelon method
4	Laplace transform definition	2019, 2020, 2022 theory family	B5	Definition + sufficient conditions nearby
5	Singular matrix definition	2015, 2021, 2023 theory family	A1	Definition + determinant test
6	Eigenvalue/eigenvector definition	2016, 2017, 2018 theory family	A3	Definition + characteristic route
7	Symmetric + skew decomposition proof	2017, 2018	A1	Formula + uniqueness outline
8	Piecewise/direct line integral family	2016, 2017	B8	Both paths correctly parameterized
9	Work on $x^2=4y$, $3x^3=8z$ curve	2020, 2024	B8	Parameterize in x and integrate cleanly
10	Cylinder surface flux family	2018, 2023	B8	Normal + cylindrical surface element
11	Volume div-F cuboid family	2022, 2023	B8	Divergence + direct triple integral
12	$\cos(at)/t$ Laplace non-existence theory	2015, 2024	B5	Limit/condition explanation

HOW TO USE THIS PAGE R1 questions should be executable from a blank sheet. Do not merely recognize them while reading the solution. Recognition without reproduction is not backlog-ready.

Same-Type Mutation Guide

Learn how a setter changes the surface while preserving the method.

Template	Common mutations	Invariant method
Rank / consistency	Parameter k moved; coefficients changed; homogeneous $\leftrightarrow$ non-homogeneous	Row-reduce and compare ranks.
Inverse matrix	Matrix size/entries changed	$[A I] \rightarrow [I A^{-1}]$, but check singularity first.
Eigen / C-H	New matrix; ask largest λ , power, inverse	Characteristic polynomial $\rightarrow p(A)=0$.
Vector motion	New $r(t)$, new evaluation time	Differentiate componentwise; speed magnitude.
Direct Laplace	New polynomial/trig/exponential mix	Linearity + table + shifting/differentiation properties.
Periodic / unit step	New period and breakpoints	Rewrite intervals; apply standard periodic/shift formula.
Inverse Laplace	Factorization changes	Partial fractions / shift / convolution selection.
ODE	New constant coefficients / RHS	Transform derivatives, insert initial data, solve $Y(s)$, invert.
Beam	Support and load pattern change	Boundary conditions + unit-step load + 4th-order transform.
Directional derivative	New scalar field, point, direction vector	Compute $\nabla\phi$, normalize direction, dot product.
Surface flux	New plane/cylinder/sphere	Find normal and correct surface element / theorem.
Closed integral	New curve/field	Select Green/Stokes/Gauss from geometry.

The 5-step template card

1. Trigger: identify the question family.
2. Formula: write the governing formula before arithmetic.
3. Setup: substitute data and choose limits/parameterization.
4. Execution: calculate in small auditable steps.
5. Check: sign, boundary, rank, orientation, units or final substitution.

Theory + Proof Priority for Backlog

Short questions can rescue a set even when the long numerical part is difficult.

Tier	Theory / proof	Set	Preparation standard
Tier 1	Orthogonal matrix; $A^{-1}=A^T$; direction-cosine orthogonality proof	A1	Full
Tier 1	Every square matrix = symmetric + skew-symmetric (unique)	A1	Full proof
Tier 1	Diagonal entries of skew-symmetric matrix are zero	A1	Full proof
Tier 1	Rank / consistent system / trivial vs non-trivial	A1/A2	Definitions + conditions
Tier 1	Eigenvalue/eigenvector definition + properties	A3	Definition + 3 properties
Tier 1	Laplace definition + sufficient conditions	B5	Full theory
Tier 1	Unit step + shifting theorem	B5	Definition + formula
Tier 1	Convolution theorem statement	B6	Statement + inverse-use form
Tier 1	Physical meaning of gradient/divergence/curl	B7	2-3 lines each
Tier 1	Scalar triple product geometrical meaning	A4/B7	Volume + coplanarity
Tier 2	Vector triple-product identity / Jacobi identity	A4	Proof outline
Tier 2	Why $L\{\cos at/t\}$ does not exist	B5	Condition + $t \rightarrow 0$ argument
Tier 2	Green / Stokes / Gauss statements	B8	Statement + when to use
Tier 2	Beam boundary conditions	B6	Memorize all three support types

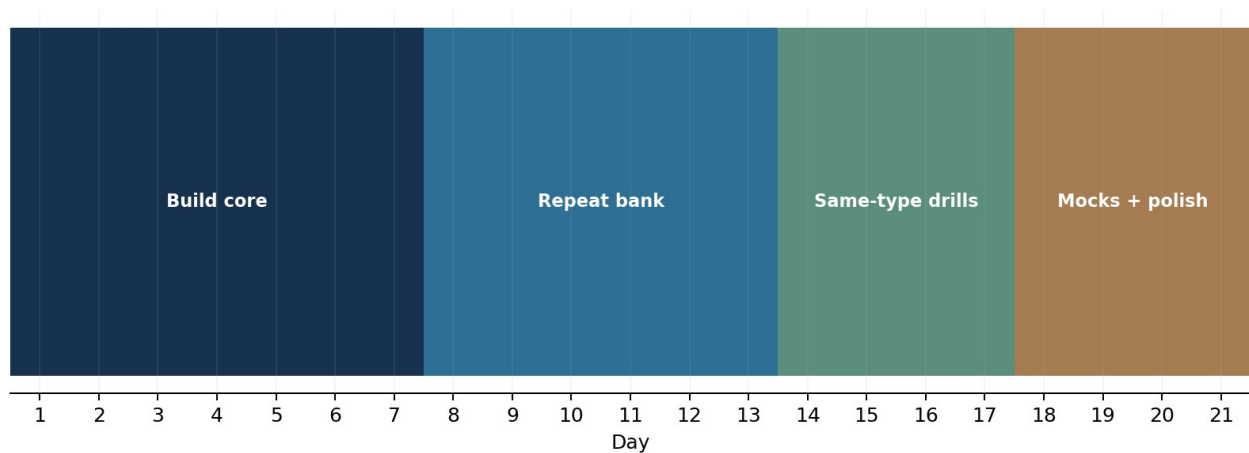
Proof-writing template

Step	What to write
1	State the defining equation / theorem condition.
2	Apply transpose, determinant, dot/cross identity, transform definition, etc.
3	Simplify one line at a time; label any key property.
4	End with the exact statement being proved.

DO NOT MEMORIZE PROSE Memorize proof skeletons and mathematical transitions. Under stress, equations are easier to reconstruct than paragraphs.

21-Day Backlog Preparation Plan

Designed for incomplete preparation and limited confidence.



Day	Main target	Output before sleeping
1	A1 definitions + symmetric/skew proofs	10 definitions from memory + 2 proofs
2	A2 rank + inverse	2 rank + 2 inverse problems
3	A2 consistency + dependence	3 parameter-system problems
4	A3 normal/echelon form	2 full reductions
5	A3 eigen + C-H	2 eigen + 1 C-H problem
6	A4 motion + work	2 motion + 2 work questions
7	Section A timed mixed test	Attempt 3 sets; mark weak steps
8	B5 direct Laplace	15 transform drill items
9	B5 periodic + unit step	Periodic repeat + 3 shifted functions
10	B6 inverse + convolution	4 inverse + 2 convolution
11	B6 ODE / integral equation	3 IVP + 1 integral equation
12	B6 beam	2 beam families + boundary-condition recall
13	B7 grad/div/curl + tangent	6 short problems
14	B8 line integrals	3 direct + 2 Green
15	B8 surface integrals	3 flux / Stokes / Gauss problems
16	B8 volume integrals	3 volume setups + 2 full evaluations
17	Exact-repeat bank	Reproduce top 12 without looking
18	Same-type mutation drill	8 changed-data variants
19	Full mock 1	3 sets A + 3 sets B under time
20	Full mock 2 + error repair	Repeat only weak templates
21	Light revision	Formula/theory cards + selection strategy + sleep

14-Day Plan

When you have two weeks, cut breadth - not repetition quality.

Day	Core work
1	A1 full + theory bank
2	A2 rank/inverse
3	A2 consistency + A3 normal form
4	A3 eigen/C-H + A4 rescue formulas
5	Section A repeat bank + timed 3-set drill
6	B5 direct + periodic
7	B5 unit step + B6 inverse/convolution
8	B6 ODE + beam
9	B7 full rescue set
10	B8 line + Green
11	B8 surface + Gauss/Stokes
12	B8 volume + exact repeats
13	Full mock + repair
14	Second mock sections + final theory/formula sweep

14-day deletion rule

- Do not start rare reference-book problems unless all R1 and R2 material is stable.
- Do not spend an hour polishing one difficult proof while core numerical templates remain weak.
- Do not reread notes passively; every session must end with written output.

EMERGENCY SCHEDULE

7-Day Backlog Rescue Plan

For survival: maximize familiar marks and reduce blank-set risk.

Day	Morning / first block	Evening / second block
1	A1 theory + repeated proofs	A2 rank/inverse
2	A2 consistency	A3 eigen/C-H
3	A3 normal form + A4 motion/work rescue	Section A 3-set timed selection
4	B5 periodic/unit-step/direct	B6 inverse/convolution
5	B6 ODE/beam basics	B7 gradient/div/curl rescue
6	B8 repeated line/surface/volume families	B8 theorem-selection + limits
7	Top exact repeats + theory cards	One short mock + final error list; stop heavy study early

7-DAY RULE You are not trying to “finish the syllabus.” You are trying to build six answerable sets plus two rescue options. Every hour must improve that outcome.

Emergency minimum

- Section A: A1 + A2 + A3 must be attemptable.
- Section B: B5 + B8 must be attemptable; choose B6 or B7 as the third.
- Memorize Tier-1 theory list completely.
- Reproduce the top 12 repeat bank once on the final two days.

3-Hour and 6-Hour Study Routines

Backlog progress requires written practice, not reading time.

If you have 3 hours/day

Block	Time	Task
Recall	20 min	Definitions/formulas from blank paper
Core problem	70 min	One set family - 2 to 4 full questions
Repeat bank	35 min	R1 reproduction
Mutation drill	35 min	One R2 variation
Error log	20 min	Rewrite mistakes + tomorrow targets

If you have 6 hours/day

Block	Time	Task
Recall warm-up	30 min	Formula/theory cards
Set 1	90 min	Core numerical set
Set 2	90 min	Second core set
Repeat bank	60 min	R1 timed reproduction
Same-type drill	60 min	Changed-data questions
Review	30 min	Error log + weak theory

The error log - mandatory columns

Question	Error type	Why it happened	Correct trigger	Re-test date
Example	Wrong surface normal	Skipped geometry check	Find ∇g and orientation before integration	Tomorrow

Mock Exams That Actually Improve Backlog Performance

Do not use mocks only to measure yourself - use them to change selection behavior.

Three mock levels

Level	Format	Purpose
Micro mock	1 set, 30-35 min	Check whether a set is truly exam-ready.
Section mock	4 sets shown; choose and answer 3	Train selection under uncertainty.
Full mock	Both sections; 3+3 answer rule	Train time, switching cost and fatigue.

Set readiness score

Criterion	0	1	2
Recognition	Do not know method	Know method after thinking	Recognize immediately
Setup	Need notes	Minor hesitation	Correct setup from memory
Execution	Cannot finish	Finish with errors	Finish cleanly in time
Theory backup	None	Some definitions	Full short-question coverage

Score each set out of 8. A core set should reach 7-8 before the exam. An insurance set can be 5-6 if its short theory and first steps are reliable.

After every mock

1. Circle every question you should have skipped.
2. Mark time lost to arithmetic versus method confusion.
3. Rewrite only the wrong step - then re-solve the full question the next day.
4. Update your 3+1 set selection if one insurance set becomes stronger than a core set.

3-of-4 Selection and Time Control

The backlog paper is not the place to prove bravery.

First 8 minutes: scan, classify, choose

Mark	Meaning	Action
G - Green	Exact repeat / fully familiar	Choose first
Y - Yellow	Same type, but arithmetic or wording changed	Choose if method is clear
R - Red	New, unclear, or time-heavy	Avoid unless needed

Selection algorithm

1. In each section, identify at least 2 Green sets.
2. Choose the third set from Green/Yellow based on method certainty, not topic preference.
3. Keep the fourth set as a fallback until you confirm no hidden difficulty in your chosen three.
4. Start with the highest-confidence set to bank marks and calm arithmetic.
5. If stuck for 6-8 minutes with no setup progress, move to the next sub-question or set.

Answer presentation for partial credit

- Write the theorem/formula before substitution.
- Show matrix operations line by line.
- Draw a small region/support sketch when limits or beam conditions matter.
- Box or underline the final result.
- If you cannot finish, leave a correct setup rather than an empty space.

DO NOT OVERCOMMIT A backlog candidate loses more from spending 35 minutes on one bad question than from leaving a difficult sub-part after writing the correct setup. Protect the six-set plan.

Mistakes That Turn Familiar Questions into Lost Marks

Most backlog damage is procedural, not conceptual.

Failure mode	Why it happens	Prevention
Memorized answer but changed data unnoticed	Exact-repeat bias	Underline every changed coefficient, limit, breakpoint and initial condition.
Using inverse for a singular matrix	Autopilot	Check det/rank before claiming A^{-1} exists.
Wrong rank conclusion	Skipping augmented matrix	Always write rank(A) and rank([A B]) explicitly.
Wrong eigen sign	Characteristic determinant expansion error	Use trace/determinant sanity checks.
Unit-step shift not rewritten in (t-a)	Pattern memory without algebra	Convert $g(t)$ to $g((t-a)+a)$ before transform.
Beam boundary conditions mixed up	Support types not memorized	Use a 3-row support-condition card daily.
Missing Jacobian r	Rushing cylindrical triple integral	Write dV first.
Wrong normal/orientation	Diagram skipped	Sketch and label outward/positive direction.
Passive revision	Feels productive	End every study block with a closed-book written test.

Last 24 Hours

No new chapters. Stabilize the marks you already built.

Do

- Reproduce the exact-repeat master bank without solutions.
- Read only your error log and Tier-1 theory/proof cards.
- Do one 20-minute set-selection drill from a past paper - do not complete the whole paper.
- Check beam boundary conditions, periodic formula, unit-step shift, consistency ranks, theorem-selection rules.
- Prepare stationery, sleep, and stop heavy calculation early enough to protect concentration.

Do not

- Open a new reference-book chapter.
- Attempt the hardest unsolved problem “for confidence.”
- Memorize final numerical answers without the method.
- Change all your core sets because of one bad practice question.

FINAL RECALL CARD A1 definitions/proofs • A2 rank/consistency • A3 eigen/C-H • A4 v, a, work • B5 periodic/unit-step • B6 inverse/convolution/beam BC • B7 grad/div/curl • B8 Green/Stokes/Gauss + limits

Backlog Priority Matrix

Use this as the final revision checklist.

Set	R1 repeat bank	R2 template bank	R3 fundamentals	Ready?
A1	Orthogonal; definitions; decomposition	Property/proof variants	All matrix-type definitions	☐
A2	Rank definition families	Inverse; consistency; dependence	Elementary operations	☐
A3	Eigen definition families	Normal form; eigen; C-H	Characteristic polynomial	☐
A4	Triple-product theory	Motion; work; geometry	Vector products	☐
B5	Periodic 3t/6; Laplace definition	Piecewise; unit step; shift	Transform table	☐
B6	Convolution/theory families	Inverse; ODE; beam	Derivative/integral properties	☐
B7	Physical meaning theory	Directional derivative; tangent; solenoidal/irrotational	∇ operator	☐
B8	Repeated line/surface/volume families	Green/Stokes/Gauss variants	Parameterization + limits	☐

Readiness thresholds

Status	Definition
CORE	Can score 7-8/8 on the readiness rubric and finish a set under time.
INSURANCE	Can score 5-6/8; theory + setup are safe even if the long calculation varies.
NOT READY	Recognition or setup still depends on looking at notes.

FINAL PRINCIPLE Backlog preparation is an optimization problem: secure the repeated and template marks first, keep one insurance set per section, and use fundamentals to survive the truly regular/new 33%.

What This Guide Uses

Transparent separation between source evidence and planning assumption.

Uploaded source evidence used

Source	How it informed the guide
Math 2123 Term Questions (2015-2025)	Set/topic history and question-style context.
KUET Math-III Theory & Proofs (2015-2024)	Repeated definitions, theory and proof families.
KUET Vector Integration Solutions (2015-2024)	Exact/near-exact line, surface and volume repeats and theorem patterns.
KUET Unit Step Function Guide (2015-2024)	Repeated periodic/unit-step question families.
KUET Beam Deflection Manual (2015-2024)	Four beam appearances, support types and boundary-condition patterns.
Lecture notes + reference books	Formula/method reinforcement and terminology.

Planning assumption supplied by the student

33% / 34% / 33% The exact-repeat / same-type / regular split is treated as a personal backlog model. It is not presented here as a measured historical frequency because the supplied archive is primarily regular-exam material rather than a dedicated backlog-paper archive.

Recommended next step

Print or annotate this guide. Mark each set as CORE / INSURANCE / NOT READY, then begin with the 21-day, 14-day or 7-day schedule that matches your remaining time. Do not change schedules mid-cycle unless your mock-readiness scores clearly justify it.

Made by 2223033

MATH 2123 • COMPLETE BACKLOG PREPARATION GUIDE